CURRICULUM VITAE

Joseph Bolton

PERSONAL DETAILS

ID number: 8707245192085

Age: 32

Gender: Male

Language: English

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Nationality: South African

QUALIFICATIONS

Current part time study: **BSc(Hons) in Statistical Sciences**, Mathematical Statistics stream. University of Cape Town [UCT] (2019 - present)

BSc Mathematical Statistics: University of Cape Town [UCT] (2015-2017)

- Golden Key Member (top 15% of stream)

Performers' Diploma in Music – Jazz: University of Cape Town [UCT] (2010-2012)

- Dean's Merit List 2010, 2011, 2012
- Diploma Awarded with Distinction in Jazz Ensemble and Piano

Bachelor of Music in Classical Composition: University of Cape Town [UCT] (2006-2009)

- Dean's Merit List 2006, 2007, 2008
- Peter Klatzow Prize for Composition 2008

High School: Rondebosch Boys' High School (2001-2005)

- A-aggregate for matric (2005)
- Emilio Farella Performance Trophy (2005)
- Augusta Reynolds Award: Best Piano Sonata Entrant, Cape Town Eisteddfod (2004)

WORK EXPERIENCE

- Statistician/Data Scientist at Eighty20 (2017-present)
  Content of previous projects:
  - Predictive modelling
  - Time-series forecasting
  - Automated customer selection through uplift modelling (modelling of heterogeneous treatment effects)
  - Mathematical optimisation
  - Exploratory data analysis

    - Data visualisation
    - R on-boarding (training) of new employees
    - Data wrangling in SQL
    - Product personalisation-engine design
    - Measuring product associations and other patterns in transaction data
    - Experimental design & hypothesis testing
    - Process automation
    - Reporting and presentation to clients
    - Creation of front-end dashboards for clients to interact directly with data
    - Use of geo co-ordinates to enable unbiased survey sampling from informal settlements (including converting image data into geo-data).

- Internship at Eighty20 (2016-2017)
    - Text mining and REGEX
    - Data wrangling
    - Process automation
- Vacation internship at Arch Actuarial (2017)
    - Reviewing financial models
- Vacation internship at Momentum (2017)
    - Excel and VBA
- Author of the music chapters in *Study and Master: Creative Arts textbooks*, Grade 7, 8, 9 – published by Cambridge University Press; over 200,000 copies sold (2014-2018)
- Over-writing/editing music content in various Cambridge University Press publications (2012)
- Composer and Producer of Educational CD, *Study and Master: Creative Arts textbooks*, Grade 7, 8, 9 – published by Cambridge University Press (2012-2013)

COMMUNITY INVOLVEMENT

- Member of the Philharmonia Choir of Cape Town (2016) – tenor
- Accompanying student music exam performances (2006-2014; over 50 students assisted)
- Chairperson, UCT Capoeira Society (2011)
- Capoeira teaching, Bonnytoun Orphanages, Western Cape (2010)

SOFTWARE EXPERTISE

- R (Statistics) and R Shiny
- H2o Machine Learning
- SQL Vertica
- Python
- Microsoft Excel & VBA
- GIT
- Photoshop

PRIMARY STATISTICAL INTERESTS

- Frequentist and Bayesian Inferential Modelling
- Uplift Modelling (estimation of heterogeneous treatment effects)
- Data Visualisation
- Multivariate Analysis
- Optimisation
- Time series modelling and forecasting
- Predictive modelling
- Deep learning
- Combinatorics

INTELLECTUAL COMPETENCIES

- Ability to analyse and synthesise; logical reasoning
- Problem-solving, mathematical/statistical skills
- Self-driven and self-teaching ability
- High-level networking skills
- High-level ability to interact with others
- Leadership
- Collaborative team-member
- Ability to innovate, conceptualise, visualise

REFERENCES

Line manager at Eighty20: Dino Bernicchi
072 213 2592
dinobernicchi@gmail.com

University statistics lecturer: Şebnem Er
072 423 8876
sebnem.er@uct.ac.za